

Name: _____



Place value in decimals

Task A: Exponential and fractional forms for decimals

1) Fill in the gaps so that each column shows a different way of writing the same value.

In fractional form		$\frac{1}{100}$		$\frac{1}{10\ 000}$
In decimal form		0.01	0.001	
In exponent form		$\frac{1}{10^2}$		
In words	one tenth	one hundredth		

2) Write the following in decimal form:

a) $2 \times 10^3 + 3 \times 10^2 + 4 + 9 \times \frac{1}{10} =$

b) $8 \times 10^3 + 4 + 0.7 + 0.02 =$

c) $9 \times 10^2 + 6 + 0.8 + 7 \times \frac{1}{10^3} =$

d) $2 \times 10^3 + 40 + 0.003 + 4 \times \frac{1}{10} + 6 \times \frac{1}{10^3} =$

Task B: Different representations of decimals

1) Complete the charts to make the decimals.

100	500	700
10	50	70
1	5	7
0.1	0.5	0.7
0.01	0.05	0.07

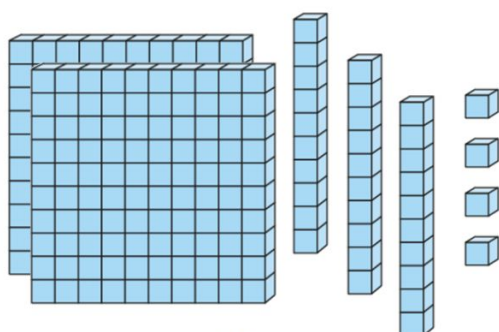
751.15

200	600
20	60
2	6
0.2	0.6
0.02	0.06

262.26

93.191

2) Three students have different numbers.



Alex

200	500
20	50
2	5
0.2	0.5

Laura

10^2	10	1	$\frac{1}{10}$	$\frac{1}{10^2}$

Sofia

Insert all the digits 6, 5, 4 and 2 into Sofia's place value chart to make:

a) a decimal greater than Laura's.

b) a decimal less than Alex's.

c) a decimal in between Alex's number and Laura's number.

Try to find more than one answer for each.

